ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE F · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice f.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers and wording. Symbols and circuits may be drawn in different but correct ways.

QUESTION 1

1.1 Any FOUR ✓✓✓✓: belts and pulleys; chains and sprockets; gears; shafts and couplings; lead screws; rack and pinion; cams. (4)

1.2 $\text{Gear ratio} = \text{driven teeth} \div \text{driver teeth}$ ✓ $= 60 \div 20$ ✓ $= 3:1$ ✓; $N_{driven} = 900 \div 3$ ✓ $= 300$ rpm ✓ (5)

1.3 Open loop: no feedback – the output is not measured, so the system cannot correct itself ✓✓ – e.g. a toaster on a timer, a fan switch ✓. Closed loop: the output is measured and fed back to the controller, which corrects any error ✓✓ – e.g. a thermostat-controlled oven, a servo motor ✓. (6)

1.4 A device that changes one form of energy (or physical quantity) into another, usually into an electrical signal ✓✓✓. Any TWO ✓✓: microphone; thermocouple; strain gauge; loudspeaker; LDR; piezo sensor. (5)

[20]

QUESTION 2

2.1 Correct OR symbol labelled S1, S2 and E1 ✓; truth table ✓✓✓✓: (5)

S1	S2	E1
0	0	0
0	1	1
1	0	1
1	1	1

2.2 Two thin metal reeds are sealed in a glass tube ✓; when a magnet comes near, the reeds are magnetised and pull together, closing the contact ✓; when the magnet moves away they spring apart ✓. Application ✓: door / window alarm, cylinder end-position sensing, counting. (4)

2.3 An NO contact is open when not operated and closes when operated ✓✓; an NC contact is closed when not operated and opens when operated ✓✓. (4)

2.4 In order ✓✓✓✓✓: identify / define the problem; investigate and gather information; develop possible solutions (design); choose and make / implement the best solution; test and evaluate (and improve). (5)

2.5 Computer Numerical Control ✓; Computer-Aided Design ✓ (2)

[20]

QUESTION 3

3.1 Any FOUR ✓✓✓✓: compressor; air receiver (tank); aftercooler / air dryer; pressure switch / safety valve; pipework (main line with drains); service unit. (4)

3.2 Air supply (compressor / supply symbol) ✓; 3/2-way valve with push-button actuator ✓ and spring return ✓; ports correctly connected (supply to P, A to cylinder, exhaust R) ✓; single-acting cylinder with spring return ✓; neat ✓ (6)

3.3 It gives an output only when air is applied to BOTH inputs (AND function) ✓✓. Application ✓✓: two-hand safety control of a press, or only allowing a cylinder to move when a guard is closed AND a start button is pressed. (4)

3.4 Two squares (2 positions) ✓ with 5 ports correctly shown ✓; flow arrows correct in both positions ✓; pilot (dashed line / triangle) actuators on both sides ✓ (4)

3.5 On the return stroke the air pushes on the rod side, where the piston rod takes up part of the area ✓; the effective area is smaller, so with the same pressure the force is smaller ($F = p \times A$) ✓. (2)

[20]

QUESTION 4

4.1 Any FOUR ✓✓✓✓: fuse; circuit breaker; overload relay; earth leakage unit (residual current device); motor protection circuit breaker; thermal cut-out. (4)

4.2.1 It is the holding (self-latching) contact ✓ that keeps K1 energised after START is released ✓. (2)

4.2.2 The circuit to K1 is broken ✓, so K1 drops out, the motor stops and H1 goes off ✓. (2)

4.2.3 H1 (green) shows that the motor is running ✓; H2 (red) shows that the overload has tripped ✓. (2)

4.2.4 When START is pressed, current flows through the E-stop, STOP and overload contacts to K1 ✓; K1 energises ✓, its holding contact closes so K1 stays on when START is released ✓; another K1 contact switches on the motor and the green lamp H1 ✓; when STOP is pressed, the circuit to K1 opens ✓, K1 drops out, the holding contact opens and the motor and H1 switch off ✓. (6)

4.3 A voltage low enough not to cause a dangerous electric shock (normally below 50 V AC / 120 V DC, e.g. 24 V) ✓✓; obtained with a safety isolating transformer that steps the mains down ✓, followed by a rectifier / power supply to give 24 V DC ✓. (4)

[20]

QUESTION 5

5.1 Any THREE ✓✓✓: loose or damaged fittings; cracked or cut tubing; worn seals in cylinders or valves; open drain valves; damaged service unit bowls; poorly cut tube ends. (3)

5.2 It gives a logical, step-by-step order of tests and decisions ✓ so that a fault is found quickly and safely without guessing ✓. (2)

5.3 In a logical order ✓✓✓✓: check that the compressor is running and there is air pressure at the service unit (gauge); check the regulator setting; check that the valve is switching (and the air supply reaches it); check the tubing for blockages, kinks or leaks; check the cylinder for leaking seals or a jammed rod. (4)

5.4 Any FIVE in order ✓✓✓✓✓: inform the people affected; switch off the machine; isolate the electrical and air supplies; fit your own padlock to the isolator; attach a danger tag with your name and the date; release stored energy (air pressure, springs, raised loads); test that the machine cannot start; only you remove your lock after the work is complete. (5)

5.5 Sequential Function Chart ✓. Drawing: initial step (double box) ✓; transition "START" ✓; step 1 with action A+ ✓; transition "a1 (A extended)" ✓; step 2 with action A− and a transition "a0" back to the initial step ✓. (6)

[20]

TOTAL: 100

END OF MARKING GUIDELINE